

SANTOSH KUMAR CHAUDHARY

Electrical and Electronics Engineer | Power Systems, Protection and Intelligent Automation
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RESEARCH INTERESTS

Power-system protection and fault diagnosis, machine learning and signal processing for power systems, smart grids and substation automation, industrial control, renewable energy integration.

EDUCATION

Bachelor of Electrical and Electronics Engineering | Nepal Engineering College (Pokhara University), **Feb 2022 – Mar 2026**
Bhaktapur, Nepal

- CGPA: 2.98/4.00; 136 credits.
- Relevant coursework: Power System Analysis, Transmission and Distribution, Switchgear and Protection, Control Systems, High Voltage Engineering, Artificial Neural Network, and Research Methodology.

PUBLICATIONS

1. **S. K. Chaudhary**, S. Khanal, P. K. Chaudhary, B. Pancha, and J. K. Yadav, “**Multi-Class Fault Type Classification Using Sequence Components and Wavelet Features with Comparative Study of Machine Learning Models and Neural Networks**,” *IEEE International Conference on ICT and Photonics (ICTP 2026)*. [DOI](#)
 - First author and oral presenter.
 - Led MATLAB/Simulink modelling, generation of a 36,000-sample dataset, sequence-component and db4 wavelet feature extraction, and ML/ANN model training and evaluation.
 - Result: Random Forest achieved 97.58% classification accuracy.
2. P. K. Chaudhary, **S. K. Chaudhary**, U. K. Karn, D. Neupane, R. K. Yadav, D. P. Guragain, and B. Shrestha, “**An Efficient Deep Learning Based Mobile Application for Pigeon Pea Disease Classification**,” *IEEE International Conference on ICT and Photonics (ICTP 2026)*. [DOI](#)
 - Second author, the paper received the Best Paper Award in Track 5: Recent Trends in Computing.
 - Developed the Android application, integrated the TensorFlow Lite model, performed deployment testing and result analysis, contributed to manuscript preparation, and supported team coordination.

RESEARCH EXPERIENCE

Wavelet and ANN-Based Fault Detection and Communication-Controlled System | Final-Year **2025 – 2026**
Project, Lead Researcher and Developer

- Designed a MATLAB/Simulink power-system protection framework using sequence components, Discrete Wavelet Transform features, and ANN/ML classifiers for multi-class fault detection and classification.
- Independently developed and fully tested a bidirectional loop linking MATLAB/Simulink, Firebase Realtime Database, and an Android application for fault-status monitoring and remote simulated circuit-breaker control.
- Validated the complete workflow using simulated power-system data. Technical implementation is complete. A journal manuscript is planned, followed by validation using data from NEA Lahan 132 kV Substation, subject to formal authorization and data-sharing requirements.

RESEARCH IN PROGRESS

Comparative Wavelet Study for ANN-Based Fault Classification | Independent Research **2026 – Present**

- Comparing wavelet families, decomposition levels and feature combinations to evaluate classification robustness and computational trade-offs. MATLAB/Simulink dataset generation is underway.

PROFESSIONAL EXPERIENCE

Electrical and Electronics Engineer | M And MB Soft Tech Pvt. Ltd., Nepal **Mar 2026 – Jul 2026**

- Contributed to electrical design and implementation of an in-house data centre, including load calculations, UPS/inverter systems, earthing, distribution boards, server-rack power, safety checks, and technical documentation.
- Tested automatic source transfer, checked ATS input/output conditions, and supported troubleshooting using AutoCAD, Excel, single-line diagrams, multimeter, clamp meter, and insulation/earthing resistance testers.

Industrial Trainee | Nepal Electricity Authority (NEA) - Lahan 132 kV Substation and Lahan Fuse Sewa **Sep 2025 – Oct 2025**

- Gained practical exposure to transformers, circuit breakers and isolators, protection panels, control-room monitoring, station battery/DC systems, switchyard equipment, meter/load recording, maintenance support, and permit-to-work safety procedures.
- Joined distribution-line patrols to locate faults and assist with consumer complaints, fuse replacement, service-line faults, low-voltage troubleshooting, meter-related issues, and supply restoration.

TECHNICAL TRAINING

- PLC Training** | Department of Electrical and Electronics Engineering, Nepal Engineering College **Dec 2025 – Feb 2026**
- Completed 21 credit hours in PLC hardware, Siemens TIA Portal, ladder-logic programming, HMI configuration, and industrial automation.
 - Implemented motor-control and traffic-light systems with live HMI monitoring on a physical PLC.

SELECTED PROJECTS AND IMPACT

- Earnify - Android Reward Application** | Founder and Android Developer **Feb 2022 – Present**
- Developed, published on [Google Play](#) in March 2025, and actively maintain a Firebase connected Android application serving approximately 1,900 registered users and 500-1,000 daily active users based on internal analytics.
 - Implemented authentication, wallet/coin data management, Firebase Cloud Messaging, dashboard driven notifications, advertisements, and background task scheduling.
- Interactive Human Body STEM Learning Model** | Three-member team project **2024**
- Co-developed a platform concept for reselling unwanted items and products made from reused materials.
 - Contributed to concept development, physical mechanisms, documentation, and team coordination.
- Circular Marketplace Platform Concept** | Hult Prize 2024-2025, three-member team **2025**
- Co-developed platform concept for reselling unwanted items and products made from reused materials
 - Contributed to the business model, platform design, market research, pitch deck, presentation to judges, and team coordination.

LEADERSHIP AND PROFESSIONAL SERVICE

- Chairperson** | IEEE Nepal Engineering College Student Branch **2024 – 2025**
- Led and coordinated approximately 3–4 technical events and student activities, coordinated the Student Branch inauguration, and managed overall workflow and operational coordination for IEEE ICTP Conference 2025.
- Secretary** | Center for Research in Applied Science and Technology (CRAST), Nepal Engineering College **2024 – 2025**
- Organized technical events, prepared meeting documentation, coordinated students and committee members, supported research fairs and competitions, and managed schedules and logistics.
- IEEE ACEI Ambassador** | Nepal Engineering College **2024 – 2025**
- Promoted IEEE ACEI initiatives, conducted workshops and awareness activities, supported AI/computing education, and coordinated with students and faculty.
- Organizing Committee Member** | Hult Prize OnCampus Program, Nepal Engineering College **2023 – 2024**
- Coordinated participating teams and supported communication between teams and the organizing committee.

HONORS, AWARDS AND SELECTED ACTIVITIES

- Best Paper Award**
IEEE ICTP 2026, Track 5: Recent Trends in Computing.
- Oral Conference Presenter**
Presented the first-author power-system fault-classification paper at IEEE ICTP 2026.
- Consolation Prize Winner**
4th-9th group, GCP-Powered AI Solutions for Building on Aelf Challenge, StackUp x Google Cloud x aelf, 2024.
- STEAM Competition Participant**
University Students' STEAM Material Design Competition 2081, Ministry of Education, Science and Technology, Government of Nepal.
- Hult Prize Competitor**
2024-2025 OnCampus Program, Nepal Engineering College, 2025.
- Software Competition Participant**
HEX-2023, Himalaya College of Engineering, 20 July 2023.

TECHNICAL SKILLS

- Power Systems and Engineering**
MATLAB (intermediate), Simulink (advanced), ETAP, AutoCAD, power-system modelling, fault simulation, load-flow, short-circuit and arc-flash analysis, relay coordination, substation systems.
- AI, Data and Signal Processing**
Python, machine learning and ANN, TensorFlow/Keras, TensorFlow Lite, DWT and sequence-component features, feature engineering, model evaluation and confusion-matrix analysis.
- Software, Cloud and Automation**
Java, Android Studio, Firebase Authentication, Realtime Database and FCM, C/C++, Siemens TIA Portal, ladder logic and HMI (training-level), Git/GitHub.
- Research and Productivity**
LaTeX/Overleaf, IEEE formatting, BibTeX, technical writing and presentation, Microsoft Excel (advanced, including dashboards, pivot tables and basic VBA).

LANGUAGES

Nepali: Native | **Maithili:** Native | **English:** Professional working proficiency | **Hindi:** Conversational